

The Semblance of Success in Nudging Consumers to Pay Down Credit Card Debt[†]

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We test a nudge in a field experiment on credit cards. The nudge shrouds the autopay enrollment option for cardholders to automatically pay exactly the credit card minimum payment each month. After six months, the nudge decreases the fraction of cardholders who only pay exactly the minimum by 23 percent. However, the nudge does not significantly reduce credit card debt. Nudged cardholders often choose autopay amounts that are only slightly higher than the minimum payment. The nudge reduces autopay enrollment, which increases missed payments. The nudge reduces manual payments by autopay enrollees. Cardholders frequently lacking liquid cash best explains our results. (JEL C93, D12, D91, G21, G51)

Policymakers often make decisions based on observing the effects of a potential policy on short-run choices that act as proxies for longer-run outcomes

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(e.g., Athey et al. 2024). Our paper provides an example of how one “nudge” (Thaler and Sunstein 2008) policy changes enrollment choices and *proximate* outcomes without changing the more challenging *distal* economic outcome of ultimate importance to policymakers.

We study this topic in the domain of credit card policymaking. Credit card payments are often at or near the minimum due: 25 percent of payments in the United Kingdom (Financial Conduct Authority 2016a) and 29 percent in the United States (Keys and Wang 2019). A potential explanation for such payment choices is that a credit card’s minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Thaler and Sunstein 2008) that acts as a psychological default (Sakaguchi et al. 2022) or focal point.¹ There is empirical support for this explanation (e.g., Stewart 2009; Keys and Wang 2019; Medina and Negrin 2022; Sakaguchi et al. 2022).

We test whether one previously untested nudge, that reduces the salience of an option to automatically only pay the minimum payment, can cause consumers to pay more than the minimum and to reduce their credit card debt. We conduct a preregistered field experiment on 40,708 UK credit cardholders. Consistent with the minimum payment acting as a focal point, the nudge is effective in shifting payment choices away from the minimum, a *proximate* outcome. However, the nudge is ultimately ineffective in changing the amount of credit card debt accumulated, a *distal* outcome. We find that credit cardholder responses to the nudge make it ineffective. These results are consistent with these consumers lacking liquid cash to reduce their debt, and we show further evidence of this using linked bank account data.

A mechanism facilitating low credit card payments is the FinTech feature called “Autopay” in the United States, known as “Direct Debit” in the United Kingdom. Autopay is a common payment mechanism used across nonfinancial (e.g., cell phones) and financial (e.g., auto loans, mortgages) products. For credit cards, enrolling in autopay is an opt-in choice. Cardholders choosing to enroll in autopay are presented with three options: automatically paying exactly the minimum amount due each month (“Autopay Min”), automatically paying a fixed amount each month (“Autopay Fix”) where the automatic payment is the maximum of a fixed amount and the minimum due that month, and automatically paying the full balance due on the statement each month (“Autopay Full”). These three autopay options are standard in the United Kingdom and the United States.² Autopay is used by 42 percent of UK cards (Financial Conduct Authority 2016a) and 20 percent to 38 percent of US cards (Consumer Financial Protection Bureau 2021), with growing use over time. Cardholders enrolled in autopay can also make supplemental, non-autopay (“manual”) payments either online or by phone.

¹Recent research (Bartels, Herzog and Sussman, 2023; Schwartz, 2024) reveals the prevalence of credit card payments exactly at or just above the minimum is more accurately described as “targeting” behavior, where consumers exert effort to meet a threshold, rather than as “anchoring” behavior (Tversky and Kahneman, 1974), partially because there is not a mass of payments just below the minimum, which would be predicted under anchoring but not under targeting.

²The largest US credit card lenders (e.g., American Express, Chase, Citi, Capital One, Discover, US Bank, and Wells Fargo) offer these autopay options.

Consumers may enroll in autopay for convenience, providing insurance against forgetting to pay a bill and also providing flexibility on how much to pay, given uncertainty over their liquid cash. Yet, autopay means that credit cardholders no longer need to actively decide each month how much to pay, and may become inattentive to their debt (Sakaguchi et al. 2022) and procrastinate on paying it down, resulting in them paying high credit card interest costs that strain their liquid cash reserves, rather than increasing their savings or consumption. Consistent with this, persistent minimum payments and high credit card interest costs are concentrated among cardholders enrolled in Autopay Min. Seventy-five percent of consumers in “persistent credit card debt”—using a regulatory definition of making nine or more minimum payments in a year on interest-bearing cards—are enrolled in Autopay Min (Financial Conduct Authority 2016a). Consumers who switch to Autopay Min pay 20 percent more in total credit card interest and fees than if they had not switched; this is despite being more likely to incur late fees without autopay (Sakaguchi et al. 2022). The 20 percent of UK credit cards enrolled in Autopay Min account for 43 percent of total interest and fees across all UK credit cards (Sakaguchi et al. 2022). Credit cardholders enrolled in Autopay Min underestimate how long it will take to pay off credit card debt if they only pay the minimum, suggesting that their payment decisions are not well-informed, and that policies to correct this bias are ineffective in reducing debt (e.g., Adams et al. 2022b).³

In our field experiment’s treated group, we remove the minimum payment as a visible and salient focal point for cardholders enrolling in autopay at card opening. We nudge consumers by removing the explicit appearance of the Autopay Min option for the treated group. Autopay Fix and Autopay Full remain visible options for both the control and treatment groups. Autopay Min remains a feasible choice for consumers if they actively choose a low Autopay Fix amount that binds at the minimum. By shrouding the Autopay Min option, we increase the salience of the Autopay Fix option, which enables an active choice and, without offsetting consumer responses, would automatically amortize debt faster.

This field experiment is an *ex ante* test of a potential nudge that the UK consumer financial protection regulator, the Financial Conduct Authority (FCA), was considering implementing, given regulatory concerns about substantial amounts of UK credit card debt (Financial Conduct Authority 2014, 2016b).⁴ The field experiment is conducted on cardholders who have self-selected to reach the autopay enrollment web page. We measure outcomes in administrative credit card data and consumer credit reporting data. Among consumers enrolled in Autopay Min in the control

³Adams et al. (2022b) find 96 percent of survey respondents, from a sample of UK consumers enrolled in Autopay Min, underestimate the time it would take to fully pay a debt if the cardholder only paid the minimum each month. More generally, approximately half of UK credit cardholders in one survey incorrectly thought the minimum payment is the amount most people paid, when in fact only a quarter do (Financial Conduct Authority 2016b). Studies across countries show that cardholders significantly overestimate the speed with which debt is cleared (and, by implication, underestimating the interest costs) if they only pay the minimum (e.g., Lusardi and Tufano 2015; Seira, Elizondo and Laguna-Müggenburg 2017; Adams et al. 2022b).

⁴See Guttman-Kenney, Leary and Stewart (2018) and Adams et al. (2022b) for other nudges the FCA tested. There are practical limits on how many potential policies regulators can test through field experiments, especially in this domain, where estimating effects on credit card debt requires large samples and observing outcomes for several months.

group of our experiment, 69 percent only pay exactly the minimum amount, indicating a large potential scope for the nudge to increase payments.

The nudge initially appears effective. It reduces initial Autopay Min enrollment from 36.9 percent of the control group to 9.6 percent in the nudged treatment group, a 74 percent decline, and increases Autopay Fix enrollment by 73 percent. These initial changes in autopay enrollment are persistent over time. The nudge successfully decreases the probability of cardholders only paying exactly the minimum by 7 percentage points or 23 percent after seven completed cycles, spanning six months. We also conduct a field experiment of the same nudge with a second lender, but after observing similarly large initial effects on autopay enrollment, this second lender withdrew before fieldwork was complete, which prevents us from evaluating its distal effects.

The nudge does *not* change credit card debt accumulated after six months. We observe null effects, on average, on credit card debt as well as spending, total payments, and borrowing costs on the specific card in the trial and across a consumer's entire portfolio of credit cards, and consumers are no more likely to pay the full balance. Such null results are critical policy inputs (Abadie 2020), especially when the null effects on real outcomes contrast with the large effects on autopay enrollment outcomes.

We investigate the mechanisms that cause the enrollment effects of the nudge to be undone, so that the effects on economic outcomes are not statistically significant. We find that three factors explain why the nudge is ineffective. First, nudged cardholders set up fixed autopay amounts that are only modestly higher than the minimum payment due, and, in the long run, essentially no higher than the minimum payment, because the minimum payment rises mechanically as card balances rise over time. Second, nudged cardholders are less likely to enroll in autopay, causing more missed payments relative to the cardholders who are not nudged. Third, nudged cardholders enrolled in autopay substitute higher automatic payments for lower manual payments, which is consistent with having limited liquid cash available to pay more overall.

Limited liquid cash balances can best explain why consumers do not reduce their credit card debt. For a selected subsample of our field experiment, we observe daily liquid cash balances from bank account data linked to our credit card data. We use these linked data to construct a new dynamic measure of liquid cash balances—the *minimum* liquid cash balance in the 90 days before card opening. This dynamic liquid cash measure reveals that half of consumers reach effectively zero liquid cash balances and 84 percent have £500 or less; this is much higher than the 13 percent that a traditional point-in-time measure of liquid cash balances would indicate. Our new measure strongly predicts credit card payment decisions six months later. After six months, consumers with small positive minimum liquid cash balances before card opening held approximately 20 percentage points less credit card debt, as a percentage of the statement balance, than those with zero or small negative minimum liquid cash balances before card opening. Consistent with limited liquid cash balances preventing consumers from reducing their debt, heterogeneity analysis shows that low liquidity consumers appear even less responsive to the nudge than high liquidity consumers. Heterogeneity analysis does not find robust evidence that the

nudge reduces debt for low-liquidity or high-liquidity consumers; the latter group may be the subgroup of consumers most expected to be making behavioral mistakes that the nudge could debias. This is important given that Allcott et al. (2024) show that the welfare effects of nudges depend on the heterogeneity of their impacts, and List et al. (2023) show that nudges' efficiency depends on their ability to debias consumers.

Our first contribution is to the literature on credit card policymaking. We show that a previously untested nudge is ineffective at reducing debt, and reveal that consumers enrolled in autopay are not as inert as they may initially appear. A challenge for consumer protection regulators is how to reduce high-cost borrowing, such as on credit cards. Around the world, informational disclosures or nudges are ineffective at reducing credit card debt (e.g., Agarwal et al. 2015; Seira, Elizondo and Laguna-Müggenburg 2017; Adams et al. 2022b). While there is a large body of research testing a range of choice architectures attempting to increase retirement savings (see reviews by Beshears et al. 2018 and Gomes, Haliassos and Ramadorai 2021), there are fewer studies testing ways to reduce credit card debt, and they tend to be information-based interventions, such as letters, emails, and text alerts (e.g., Agarwal et al. 2015; Medina 2021; Seira, Elizondo and Laguna-Müggenburg 2017; Adams et al. 2022b) rather than more forceful changes in choice architecture, such as the nudge we test (an exception being the small pilot of Karlan and Zinman 2012).⁵

Our second contribution is to the literature on the drivers of credit card indebtedness. Credit card debt is a well-established puzzle (e.g., Zinman 2015; Gomes, Haliassos and Ramadorai 2021). Our research helps to understand some of the factors that contribute to this. Our findings show that "anchoring" to minimum payments in credit cards appears less economically important than previously thought. While the headline results for Keys and Wang (2019) and Medina and Negrin (2022) both show that changes in minimum payments cause changes to payments, consistent with anchoring, neither finds that this causes significant changes in debt.⁶ It is still possible for anchoring to be important for manual credit card payment choices, given lab evidence (e.g., Stewart 2009; Navarro-Martinez et al. 2011; Guttman-Kenney, Leary and Stewart 2018; Sakaguchi et al. 2022). However, our research shows that anchoring does not appear to be the reason why consumers enrolled in autopay do not pay more of their credit card debt. Instead, these consumers' payments are constrained by frequently lacking liquid cash balances.

Our third contribution is to the literature on nudging. Our study is an example of how important it is for policymakers to evaluate nudges on how they impact distal outcomes. Our experiment shows that policymakers should not assume that changes in enrollment choices lead to changes in economic outcomes. For example, if a policymaker only observes the effects of the nudge on the composition of autopay enrollments, it may appear effective: We estimate that it would reduce

⁵This is understandable given there is an incentive for firms to conduct field experiments to find ways to increase retirement contributions because higher assets generate higher revenues. However, it is not in the incentives of credit card lenders to test or publicize ways to reduce credit card debt, as lower debt generates lower interest revenue.

⁶Neither of these studies observe autopay enrollment or liquid cash balances, and the minimum payment also remains salient in both studies.

debt by approximately 4.5 percent. However, examining the effects of the nudge on actual debt reveals that the nudge is ultimately ineffective. We show that even when a nudge not only changes enrollment but also successfully changes *proximate* outcomes (e.g., only paying the minimum), the same nudge may still not change the more *distal* outcomes (e.g., debt), and therefore it is critical to measure effects on the latter. Without ex ante tests that measure distal outcomes, policies that sound appealing may be introduced that are costly and ineffective at changing distal outcomes, for example, as discovered ex post with the US CARD Act disclosures (Agarwal et al. 2015; Keys and Wang 2019). Across financial domains, nudges can shift enrollments, but consumers may also subtly counteract these effects. For example, Choukhmane (2025) finds the long-run effects of automatic enrollment defaults on savings are smaller than short-run contribution increases found in the earlier academic literature (e.g., Madrian and Shea 2001; Thaler and Benartzi 2004). Some nudges are still highly effective even when potential countervailing effects are measured. For example, Chetty et al. (2014) show that automatic enrollment increases retirement savings. Other nudges may be effective in changing the targeted behavior; however, they may have unintended side effects. For example, Medina (2021) shows that credit card reminders help consumers to avoid credit card late fees, but unintentionally lead them to incur more overdraft fees. Our study contributes to the broader debate on the effectiveness of nudges (e.g., Thaler 2017; Laibson 2020). DellaVigna and Linos (2022)'s meta-study documents the heterogeneous effects of nudges and provides evidence for publication bias.⁷ Recent meta-studies (e.g., Beshears and Kosowsky 2020; Saccardo et al. 2025) show that very few studies examine the long-term effects of nudges, as we do, or measure broader outcomes that could offset effects, and, when these are observed, nudges are less effective.⁸ Our nudge's contrasting proximate and distal effects are an important example to consider when designing and evaluating policies.

The paper proceeds as follows. Section I explains our field experiment, the design of the nudge, and how the field experiment is implemented. Section II describes the data we use and our empirical methodology. We present our experimental results in Section III. Section IV studies the mechanisms that explain our experimental results, including using linked bank data to study liquid cash balances. Finally, Section V concludes. Additional results are contained in the Supplemental Appendix, this paper's code and data depository (Guttman-Kenney et al. 2025), and in our earlier working paper (Guttman-Kenney et al. 2023).

⁷DellaVigna and Linos (2022) show the average effect among published academic studies of nudges is 8.7 percentage points, a 33.4 percent increase, whereas the average effects from the population of studies from Behavioral Insights Teams are smaller: 1.4 (8 percent increase). In this meta-study, the average effect is compiled from a mixture of short-run and long-run outcomes based on the outcomes observed in the original studies.

⁸Beshears and Kosowsky (2020) review 174 nudge papers and find only 17 examine long-term effects and only 12 study effects on outcomes that could offset the nudge's effectiveness. Saccardo et al. (2025) show effects of Behavioral Insights Teams' nudges are smaller when the outcomes are more broadly defined or measured over longer time horizons.

I. Field Experiment

A. Nudge Design

Credit cardholders have broad discretion in how much to pay on their credit card each month; paying any amount between the minimum due and the full balance fulfills their contractual obligations. The minimum payment due is typically calculated by $\max\{\pounds 5, 1 \text{ percent statement balance} + \text{interest} + \text{fees}\}$. This formula means that the minimum payment amount will typically decrease as balances decrease. If a cardholder is only paying the minimum, then (i) their payment is effectively only servicing debt interest payments, with interest rates near 20 percent being typical, and (ii) debt reduction only happens at all if new spending is less than 1 percent of the statement balance. Even with *no* new spending, debt paydown is only 1 percent of the statement balance per month if a cardholder only pays the minimum.

In our field experiment, we vary how autopay enrollment options are presented to UK consumers who have opened a new credit card account. We nudge autopay enrollment at card origination because these initial autopay decisions are highly persistent (e.g., Sakaguchi et al. 2022; Adams et al. 2022b; Wang 2024). When a consumer opens a new credit card online, they are typically presented with the option to enroll in autopay. If a consumer decides to opt-in, they are normally presented with three autopay options: Autopay Full, Autopay Fix, and Autopay Min. All of these autopay options are shown to our control group. At this stage, consumers can still decide against enrolling in any type of autopay by not completing the enrollment process. They could also return and complete the autopay enrollment later. The treatment is a previously untested nudge that shrouds the option to automatically make only the minimum payment each month. This is done by removing the explicit appearance of the Autopay Min option. The nudge makes it difficult for inattentive consumers to default into automatically only paying exactly the minimum each month. Removing the Autopay Min option increases the salience of the alternative Autopay Fix and Autopay Full options. Supplemental Appendix Figure A1 shows the designs of the control and treatment.

While Autopay Min is a common payment option, cardholders also have the option to enroll in an alternative autopay option that would pay down debt faster: “Autopay Fix.” Autopay Fix is calculated by $\max\{\text{Autopay Fix } \pounds, \text{Minimum Payment Due}\}$. If a consumer is enrolled in Autopay Min, their Autopay Min amount will decrease when balances decrease due to the minimum payment formula described above.⁹ Consider a typical credit card balance of $\pounds 1,000$, assuming 18.9 percent APR and no further card spending. This would take 18 years and 6 months to pay off if only exactly the minimum is paid each month, such as through Autopay Min. It takes a long time for this debt to be paid off because, as balances decrease, the minimum payment amount, and therefore the Autopay Min amount, will also decrease from around $\pounds 25$ to $\pounds 5$ over time. However, by fixing the monthly payment to $\pounds 25$ each month, the debt pay-off horizon falls to 5 years and 1 month, saving over

⁹This issue would also occur if autopay options to pay a percentage of the statement balance were provided. The effect of offering such options is not tested in our experiment and is left for future research.

£750 in interest costs. If a consumer chooses a slightly higher fixed monthly payment amount, this would further reduce the time of payoff and borrowing costs. For example, with a fixed payment of £50 each month, the debt payoff time decreases to 2 years, and the interest costs become only £191, compared to £509, if paying a fixed amount of £25 each month.

We want consumers to make active choices (e.g., Carroll et al. 2009; Keller et al. 2011) about how much they want to pay each month using the Autopay Fix option, or to target the Autopay Full option. Because few consumers can pay their credit card debt in full each month, the nudge is primarily designed to work by increasing Autopay Fix enrollment which, relative to Autopay Min, is expected to increase automatic payments, which would then be expected to reduce debt and interest costs. A further possible effect could be to increase consumer spending via debt paydown, as that increases credit limit availability (e.g., Gross and Souleles 2002; Agarwal et al. 2017).

We purposefully do not include an alternative recommended Autopay Fix amount in the treatment because we do not want to replace the Autopay Min as a focal point with another focal point (other than the Autopay Full option), distorting consumers' payment choices. This design choice is motivated by US studies (e.g., Agarwal et al. 2015; Hershfield and Roese 2015; Keys and Wang 2019) that find that providing consumers with credit card payment scenarios unintentionally reduces payments for some consumers.

While there is no longer an explicit Autopay Min option in the treatment arm, cardholders can choose an operationally equivalent option by setting an Autopay Fix of £5 (or less). These two options are equivalent, as the minimum payment is calculated as $\max\{\text{£5}, 1 \text{ percent statement balance} + \text{interest} + \text{fees}\}$ and so is greater than or equal to £5 by construction. This means that when the minimum payment due in a particular month is more than £5, the autopay attempted to be taken will adjust accordingly, regardless of whether a consumer has an Autopay Fix amount of £5 or an Autopay Min. This equivalence is not highlighted to consumers, and we do not expect them to be aware of this or work it out. We explain this to show that the treatment does not restrict consumers' choice of an autopay option to pay the minimum, and therefore the treatment is a nudge rather than a restriction (the Autopay Min option is no longer explicitly labeled on the website).

B. Experiment Implementation

We test the nudge through a randomized controlled trial (RCT) conducted in the field on UK credit cardholders. The FCA invited all UK credit card lenders to voluntarily participate in a field trial. Two lenders were willing and technically able to participate within the timelines necessary to inform FCA policymaking.

When a consumer is applying for a new credit card online and has been accepted by a lender they have the option to set up autopay on this new card. If a consumer selects the option confirming that they want to enroll in autopay, they are included in the experiment. At this point consumers are randomly assigned to either control or treatment (the nudge). Once allocated to control or treatment the consumer would view the same assigned screen if they returned to the autopay landing page within

30 days. If a consumer in either the control or treatment group phones the lender's call center, they could still enroll in an explicit Autopay Min if they ask to do so. Thirty days after card opening, cardholders in both the control and treatment groups have identical screens containing explicit Autopay Min enrollment options. This is relevant if a cardholder comes back to the autopay launch page to change their autopay enrollment status. Inclusion in the experiment is irrespective of whether the autopay enrollment process is completed after reaching the autopay enrollment screen. We carried out qualitative consumer testing to ensure consumers would understand how to navigate the treatment, the experiment was reviewed by the FCA's Institutional Review Board's governance and by both lenders' governance, and we also sought feedback from all UK credit card providers and large consumer organizations. Lenders did not report any consumer complaints regarding the trial or the lack of an explicit Autopay Min option in the treatment.

Our field experiment was conducted on two UK lenders. The main lender is a large UK firm, and our experiment included 40,708 credit cards that were newly issued between February and May 2017. We wanted at least 20,000 cards in each of the control and treatment groups to ensure we had sufficient statistical power to estimate the effects on debt. The final achieved number is slightly higher, as for logistical reasons, new cards were included until the end of May 2017. We also conducted the experiment with a second lender. The second lender stopped the experiment after one week of fieldwork due to the lender's concern over the large treatment effects on autopay enrollments. The second lender's experiment was not restarted, and the pre-agreed target sample size was not reached. The second lender's experiment achieved a sample size of 1,531 cards, which was insufficiently powered to distinguish between null results and imprecisely estimated non-null effects. Had we known this second lender would stop fieldwork prematurely, we would not have started the experiment with this lender. For completeness, the results from the second lender are in Supplemental Appendix B. The rest of this paper is based on the field experiment with the main lender unless explicitly stated otherwise.

II. Data and Methodology

A. Data

Our data is gathered by the FCA using its statutory powers (Financial Conduct Authority 2018). All analysis is conducted on anonymized data (Guttman-Kenney et al. 2025).

Credit Card Data.—We collect detailed administrative data for every credit card in the experiment. We observe data recorded at card origination (e.g., opening date, interest rates, initial credit limit) and across all statements (e.g., statement balances, spending) through December 2017. A completed statement cycle is one where the payment due date for a credit card statement has passed. For the main lender, we observe 7 completed cycles for effectively all cards (99.9 percent) and up to 11 for the cards opened at the start of the experiment's fieldwork. For the second lender, we observe 12 completed cycles. Each individual payment made

against these statements is observed including the date, amount, and whether the payment is made via autopay or manually.

Consumer Credit Reporting Data.—For consumers in the experiment, we gather consumer credit reporting data (see Gibbs et al. (2024) for a review of this data), which enables us to observe effects across the consumer's debt portfolio. Consumer credit reporting data provide monthly tradeline-level data showing credit limits, balances, payments, and arrears for each account, from card opening to the end of 2017. We observe statement balances (i.e., before payments), payments, balances after payments (i.e., debt), and indicators for whether a card only paid the minimum. UK credit reporting data contain payments data for all credit cards, something that does not occur in US credit reporting data (Guttman-Kenney and Shahidinejad 2025). We observe credit risk scores and income estimates (where available) at two points in time: the month before the card was opened and nine months afterward. These data mean that if the treatment caused a large increase in payments to credit cards in the experiment that caused financial distress elsewhere in their portfolio, we could observe it. Credit card administrative data and credit reporting data are linked using an anonymous key created for this project.

Bank Account Data.—We also observe bank account data, consisting of checking/current accounts, and savings accounts for the subset of cardholders who hold these with the main credit card lender in our experiment. The bank account data report end-of-day balances each day up to a year before the experiment started (or when the account was opened) and up to June 2017—a month after the last card is enrolled in our experiment. After restricting this data to cardholders who appear to be actively using this bank as their primary bank account, we observe 3,753 cardholders or 9.2 percent of our field experiment. Additional details are provided in Supplemental Appendix E.

Summary Statistics.—As the experiment is conducted on newly opened cards, we describe summary statistics for the control group after seven cycles. We observe a diversity of credit cardholders in our data with a wide range of interest rates, credit scores, credit card credit limits, ages, and incomes. The mean credit card statement balance after cycle seven is £2,164 and £1,963 after payments. Cardholders often hold other credit cards in their portfolio: Their mean credit card portfolio statement balances, aggregated across cards held in consumer credit reporting data, are £3,917 and £3,432 after payments. Credit card portfolio balances both before and after payments are higher than the consumers' mean income of £2,437. In line with the motivation for our experiment, the cardholders in our control group are often only paying exactly the minimum, *especially* those enrolled in Autopay Min. Thirty percent only pay the minimum payment in the seventh cycle, and 19 percent only pay the minimum six or more times in the first seven cycles. Eighteen percent had paid in full six or more times. Among consumers enrolled in Autopay Min, 69 percent only pay exactly the minimum amount, and only 20 percent pay more than 10 percentage points in excess of the minimum. For more details, see summary statistics in Supplemental Appendix Table A1 and Supplemental Appendix Figure A6.

B. Empirical Methodology

Before analyzing data, we preregistered our methodology. Our preregistration designates primary outcomes, regression specifications, and thresholds for statistical significance (Adams et al. 2022a). We structure our analysis into three parts: primary, secondary, and tertiary analyses. Conducting secondary analysis depends on the results of the primary analysis. We design and implement tertiary analysis after examining the data. This structure limits the potential for data mining or p-hacking. The primary analysis focuses on ten primary, real economic outcomes upon which the nudge's effectiveness is evaluated.

The first six primary outcomes (1 to 6) measure the impact on the credit card in the experiment ("target card"), constructed from microdata collected from the lender. All these primary outcomes are bounded between zero and one. Outcomes 1, 2, and 3 are binary: (1) any minimum payment, (2) any full payment, (3) any missed payment. Outcome (4) is a measure of credit card (revolving) debt: statement balance net of payments (percent statement balance). We examine multiple moments because credit card payments have a non-normal, bimodal distribution (e.g., Keys and Wang 2019), with the tails being economically important. Outcome (5) is a measure of borrowing costs (combining interest and fees): costs (percent statement balance). Outcome (6) is a measure of consumption: spending (percent statement balance). Our measures of debt, consumption, and costs are all normalized by dividing by statement balances, denoted by percent statement balances, to deal with fat-tailed credit card balances. Normalizing our measures of debt by credit card statement balance is not ideal, as it means our outcome is a ratio of two endogenous components. To address this, our secondary analysis also shows the numerator and denominator in levels (£) separately (and having completed the analysis, we find the results are consistent). Primary outcomes 7 to 10 are analogous to primary outcomes 1 to 4, but are constructed using consumer credit reporting data to assess the impact across a consumer's portfolio of credit cards. These primary outcomes are: (7) share of credit card portfolio only paying minimum, (8) share of credit card portfolio making full payment, (9) share of credit card portfolio missing payment, and (10) credit card portfolio balances net of payments (percent statement balances). See Supplemental Appendix A for more details on primary outcome definitions.

Following Benjamin et al. (2018), we regard a p -value of 0.005 as the threshold for statistical significance, but also highlight where results are "suggestively significant" at the 0.01 and 0.05 levels. A 0.005 significance aligns with 14+ Bayes factors—often considered substantial evidence for a hypothesis. This approach is analogous to applying Bonferroni or family-wise error corrections to ten outcomes evaluated at the 0.05 significance level.

The preregistered secondary analysis considers a broader set of outcomes and empirical approaches to understand our results and their robustness. This secondary analysis measures the effects of the nudge on autopay enrollment and uses pounds (£) amounts of credit card debt and payments as robustness checks of our primary outcomes.

We can causally identify the effects of the treatment on consumers in our field experiment through our randomization of a consumer to the control or the

treatment.¹⁰ Equation (1) shows the OLS regression specification used to derive average treatment effects. To estimate this, we construct an unbalanced panel with one observation for each consumer's (i) credit card statement cycle (t) observed. This panel is unbalanced, as some cards are opened earlier than others. In this specification, δ_τ are the coefficients on the interaction terms between treatment indicator and statement-cycle indicators. Therefore, δ_τ shows the average treatment effect for $\tau \in \{1, 2, \dots, T\}$ cycles since the beginning of the experiment. Our regression includes a constant (α), a vector of time-invariant control variables ($\mathbf{X}_i'$) constructed using information on the new credit card opened, and cardholder data from before the start of the experiment (all controls are listed in footnote 11).¹¹ We also include time fixed effects: We control for both the statement cycle (γ_t) and year-month ($\gamma_{m(i,t)}$) because statement cycles do not perfectly align with calendar months and new credit cards have different opening dates. Standard errors are clustered at the consumer level.

$$(1) \quad Y_{i,t} = \alpha + \sum_{\tau=1}^T \delta_\tau (TREATMENT_i \times CYCLE_\tau) + \mathbf{X}_i' \beta + \gamma_{m(i,t)} + \gamma_t + \varepsilon_{i,t}.$$

For our primary analysis, we focus on the outcomes at the seventh completed statement cycle (δ_7), as this is the last cycle where the panel is “balanced,” i.e., where we observe a completed payment cycle for all cards. The seventh cycle is complete when its due date for payment has passed: a mean of 195 and a median of 196 days from card opening, with a range of 175 to 245 days. Seven cycles are six genuine cycles over six months, as a new card's first statement is typically less than a month—in our data, the first statement is issued a mean of 12 and a median of 11 days from card opening—to onboard the card onto a particular billing cycle, and this first statement has a zero payment due that makes it uninteresting, but we show for completeness. A consumer's first full cycle is statement two—the second statement is issued at a mean of 43 and a median of 42 days from card opening, when the cardholder has at least one month to view the control or treatment screens and to use their card (and, if used, has a nonzero payment due).

In tertiary analysis, we pool data across all cycles to provide more statistical power. We modify equation (1), replacing the dynamic $TREATMENT_i \times CYCLE_\tau$ with the static $TREATMENT_i$ shown in equation (2), where our single static parameter of interest is δ . When interpreting δ , it is important to note that this may provide

¹⁰ Supplemental Appendix Table A3 shows that allocation to the treatment group is balanced, on average, across measures. However, we observe the probability of being in the treatment group slightly varies with credit card limit. Investigating this revealed that the “live” randomization code used by the lender was not completely random: 526 more consumers (0.65 percent) are allocated to control than to treatment. As consumers applying for credit cards were unaware of (and unable to manipulate) their probability of being allocated to treatment, we can recover balance between treatment and control through conditioning on covariates. Conditioning on observables using our preregistered controls does not change our results; see our earlier working paper (Guttman-Kenney et al. 2023).

¹¹ The controls ($\mathbf{X}_i'$) are gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card purchases rate, subprime credit card, Any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, Any credit card secondary cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (credit score, DTI ratio, and any mortgage debt), which are from the month preceding card origination. For outcomes constructed from consumer credit reporting data, up to eleven dummies for lags of outcomes are included as controls ($\mathbf{X}_i'$) for months preceding the start of the experiment.

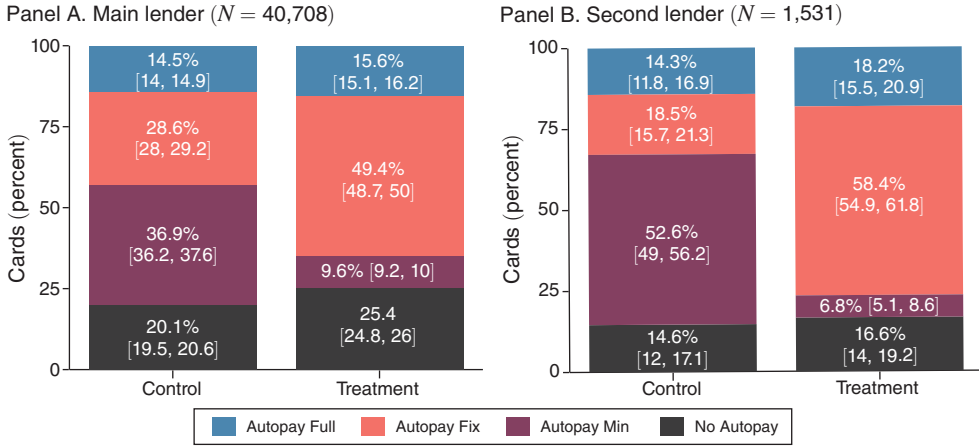


FIGURE 1. AUTOPAY ENROLLMENT FOR CONTROL AND TREATMENT GROUPS AFTER TWO STATEMENTS, SPLIT BY LENDER

Notes: Numbers display percentage of cards enrolled in each type of autopay by the second statement cycle. Ninety-five percent confidence intervals in brackets.

a misleading view of the longer-term impacts of the nudge if the effect sizes change over time.

$$(2) \quad Y_{i,t} = \alpha + \delta TREATMENT_i + \mathbf{X}_i' \boldsymbol{\beta} + \gamma_{m(i,t)} + \gamma_t + \varepsilon_{i,t}.$$

III. Experimental Results

A. Effects on Autopay Enrollment

The first effect we examine is the mechanism through which the treatment is designed to work: changing autopay enrollment choices by the time of the second credit card statement. Autopay enrollments are secondary outcomes. Figure 1, panel A, shows that the treatment causes large, significant initial effects on autopay enrollment choices. The treatment increases the fraction of cardholders who enroll in Autopay Fix by 20.9 percentage points, a 72 percent increase in the mean of the control group. For comparison, Figure 1, panel B, shows that these enrollment effects are even greater for the second lender who stopped the field experiment early: increasing the enrollment of Autopay Fix by 40 percentage points, a 216 percent increase on the control mean. The subsequent results are all based on the main lender. Almost all the mass of increased Autopay Fix enrollment is redistributed from cardholders who enroll in Autopay Min in the control group. The treatment reduces the fraction of cardholders enrolling in Autopay Min by 27.3 percentage points, a 74 percent decrease on the mean of the control group. Autopay Min is not eliminated, as consumers in the treatment group can sign up for this through other ways (e.g., telephoning the call center). The treatment causes an increase in Autopay Full enrollment of 1.2 percentage points relative to the control mean of 14.5 percent. The treatment

also causes a decline in any autopay enrollment (i.e., Autopay Full, Autopay Fix, or Autopay Min) of 5.1 percentage points from the control mean of 80.2 percent.

The Autopay Fix amounts consumers initially choose are frequently round numbers: 62 percent of Autopay Fix amounts are for £100, £50, £200, £150, £20, £30, and £25, in descending order of frequency. Very few consumers select Autopay Fix amounts of £5 or less that are mechanically identical to Autopay Min: 2.4 percent of the treatment group set an Autopay Fix of £5 or less, and effectively no cardholders who enroll in an Autopay Fix set this exactly equal to £5 in either control (0.06 percent) or treatment (0.07 percent). These are statistically significant increases relative to the 0.5 percent in the control group that we interpret as being economically small. Consumers' initial choices of Autopay Fix amounts are persistent over time. Of those in the treatment group who are enrolled in Autopay Fix at the time of their second credit card statement, 88.3 percent remain enrolled in Autopay Fix at the time of their seventh statement. Of these, 97 percent have it set for the same Autopay Fix amount, and, on average, the difference in amount is trivial, £0.78. Of the remainder, 7.0 percent are not enrolled in any Autopay, 4.4 percent in Autopay Min, and 0.3 percent in Autopay Full. Among all cardholders in the treatment group enrolled in Autopay Fix at cycle two, the mean Autopay amount is £96.85 (median £80) compared to £104.60 (median £100) at cycle seven. This indicates that cardholders who enroll in Autopay Fix later on are choosing slightly higher Autopay Fix amounts than the initial group.

Table 1 shows statistically significant effects of the treatment on autopay enrollment using our preregistered regression specification. The regression coefficients after seven cycles, δ_7 in equation (1), are consistent with the initial changes in enrollment. Autopay Min enrollment significantly decreases by 21.7 percentage points, Autopay Fix enrollment significantly increases by 16.7 percentage points, Autopay Full increases by 0.6 percentage points, which is significant at the 5 percent but not at the 0.5 percent level, and any autopay enrollment significantly declines by 4.4 percentage points. The purple coefficients in panel A of Figure 2 show the dynamic effects on Autopay Fix enrollment, δ_t in equation (1). The initial effect increasing Autopay Full enrollment attenuates over time and becomes statistically insignificant from zero. The effects on Autopay Fix and Autopay Min enrollments also attenuate over time, but these effects remain large. As the initial autopay choices in the treatment group are highly persistent, this attenuation is primarily driven by the control group "catching up" by naturally switching from Autopay Min to Autopay Fix or Autopay Full. The observed changes in autopay enrollments—the nudge making consumers more likely to choose full, less likely to choose minimum, and changing the distribution of autopay amounts—are consistent with lab evidence studying the manual payment environment (e.g., Stewart 2009; Navarro-Martinez et al. 2011; Sakaguchi et al. 2022; Guttman-Kenney 2024) that finds that the salience of the minimum payment amount distorts the control group's choices.

Which types of consumers stop enrolling in any autopay due to the nudge? We segment consumers into three groups following the methodology in Marbach and Hangartner (2020). This methodology requires the monotonicity assumption that the nudge does not make consumers more likely to enroll in any autopay. We estimate that 4.2 percent of the sample are "Compliers," who would have enrolled in

TABLE 1—AVERAGE TREATMENT EFFECTS FOR AUTOMATIC PAYMENT ENROLLMENT
OUTCOMES AFTER SEVEN STATEMENT CYCLES

Outcome	Estimate (SE)	95% C.I.	p-value	Control mean
Any autopay	−0.0437 (0.0041)	[−0.0517, −0.0356]	0.0000	0.7811
Autopay Full	0.0065 (0.0028)	[0.0009, 0.0120]	0.0217	0.1309
Autopay Fix	0.1670 (0.0045)	[0.1583, 0.1757]	0.0000	0.2955
Autopay Min	−0.2172 (0.0041)	[−0.2251, −0.2092]	0.0000	0.3547
Autopay Fix exceeding minimum payment amount	0.0859 (0.0043)	[0.0774, 0.0943]	0.0000	0.2523

Notes: The table shows average treatment effects after seven statement cycles. Each row of the table shows estimates from separate regressions with different outcomes. Estimated treatment effects are from coefficient (δ_7) on interaction terms between treatment indicator and the seventh statement cycle indicator in the OLS regression specified in equation (1). Regressions also include interactions between treatment indicator and other statement cycles, statement cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card purchases rate, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (credit score, DTI ratio and any mortgage debt), which are from the month preceding card origination. Standard errors are clustered at the consumer level. There are 40,708 credit cards with 368,162 observations.

autopay but did not do so due to the nudge, 21.9 percent are not enrolled in autopay regardless of the nudge (“Always Takers”), and 73.9 percent are enrolled in autopay regardless of the nudge (“Never Takers”). On average, compliers, compared to the other groups, have lower credit scores, lower estimated incomes, lower unsecured debt-to-income (DTI) ratios, smaller credit limits, fewer credit cards, lower debt, and are also more likely to be younger and male. See Supplemental Appendix Table A7 for details.

B. Effects on Long-Term Real Economic Outcomes

Table 2 shows the effects on our ten primary outcomes. These estimates use our preregistered regression specification and show our treatment estimates seven cycles after card opening, the δ_7 coefficient in equation (1).

Only Paying the Minimum.—We find a large and persistent effect of the nudge, making cardholders significantly less likely to only pay exactly the minimum. Table 2 shows that the nudge causes a significant decrease in the probability of only paying exactly the minimum after seven cycles of 7.1 percentage points, with a 95 percent confidence interval of −6.2 to −7.9 percentage points. This is a 23 percent decrease relative to the control group mean. Figure 2, panel B, shows the effect is −10.9 percentage points in the second cycle and stabilizes near −7 percentage points by the

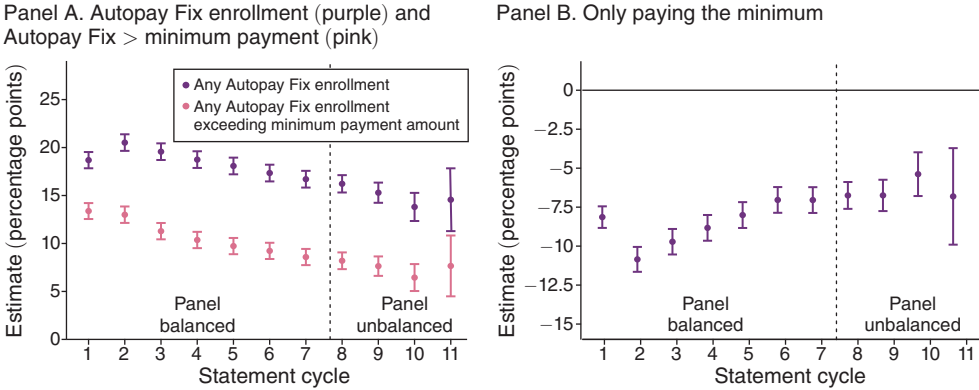


FIGURE 2. AVERAGE TREATMENT EFFECTS ON (PANEL A) AUTOPAY FIX ENROLLMENT, AND (PANEL B) ONLY PAYING THE MINIMUM, ACROSS 1–11 STATEMENT CYCLES

Notes: Treatment effects from coefficients (δ_x) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in equation (1). Regression outcomes in panel A are any automatic fixed payment enrollment (pink) and any automatic fixed payment enrollment where the fixed amount is not binding at the minimum payment amount (purple), and in panel B the outcome is only paying exactly the minimum payment. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card purchases rate, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (credit score, DTI ratio, and any mortgage debt), which are from the month preceding card origination. Error bars are 95 percent confidence intervals. Standard errors are clustered at the consumer level. There are 40,708 credit cards with 368,162 observations.

sixth cycle. The effect of the nudge on only paying the minimum is smaller than the effect size on Autopay Min enrollment shown in the previous subsection. This is because cardholders enrolled in Autopay Min can also make additional manual payments to pay more than the minimum. Also, as some cards have no balance due, and therefore no minimum payment and no payments taken, we classify these as full payments. The average treatment effect on the share of a cardholder’s credit card portfolio that only pays the minimum, constructed from consumer credit reporting data, is a third of the size for the card for which the treatment is targeted. This smaller overall effect across the credit card portfolio is due to consumers holding multiple cards, only one of which is directly affected by the nudge.

We profile consumers who the nudge makes less likely to pay only exactly the minimum, again following the methodology in Marbach and Hangartner (2020) used in the previous subsection, but now with the monotonicity assumption that the nudge does not make consumers more likely to pay only exactly the minimum. “Compliers,” who would have only paid the minimum but did not do so because of the nudge, make up 6.9 percent of the sample; 69.9 percent do not pay only the minimum regardless of the nudge (“Always Takers”); and 23.2 percent pay only the minimum irrespective of the nudge (“Never Takers”). Compliers, on average, have similar characteristics to other groups. See Supplemental Appendix Table A8 for details.

TABLE 2—AVERAGE TREATMENT EFFECTS FOR PRIMARY OUTCOMES AFTER SEVEN STATEMENT CYCLES

Outcome	Estimate (SE)	95% C.I.	<i>p</i> -value	Control mean
Any minimum payment	−0.0705 (0.0042)	[−0.0787, −0.0622]	0.0000	0.3012
Any full payment	0.0040 (0.0037)	[−0.0032, 0.0112]	0.2747	0.2397
Any missed payment	0.0038 (0.0019)	[0.0002, 0.0075]	0.0409	0.0369
Statement balance net of payments (% statement balance)	−0.0051 (0.0035)	[−0.0119, 0.0017]	0.1428	0.6936
Costs (% statement balance)	−0.0003 (0.0006)	[−0.0015, 0.0010]	0.6782	0.0111
Spending (% statement balance)	0.0025 (0.0031)	[−0.0036, 0.0087]	0.4199	0.2007
Share of credit card portfolio only paying minimum	−0.0264 (0.0027)	[−0.0317, −0.0210]	0.0000	0.2012
Share of credit card portfolio making full payment	0.0011 (0.0033)	[−0.0054, 0.0076]	0.7340	0.4414
Share of credit card portfolio missing payment	−0.0000 (0.0013)	[−0.0025, 0.0024]	0.9701	0.0236
Credit card portfolio balances net of payments (% statement balances)	−0.0053 (0.0031)	[−0.0115, 0.0008]	0.0896	0.6954

Notes: The table shows average treatment effects after seven statement cycles. Each row of the table shows estimates from separate regressions with different outcomes. Estimated treatment effects from the coefficient (δ_7) on interaction terms between the treatment indicator and the seventh statement cycle indicator in the OLS regression specified in equation (1). Regressions also include: interactions between treatment indicator and other statement cycles, statement cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card purchases, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination, except for the variables constructed from consumer credit reporting data (credit score, DTI ratio, and any mortgage debt), which are from the month preceding card origination. For outcomes constructed from consumer credit reporting data up to eleven dummies for lags of outcomes are included as controls ($\mathbf{X}_i^l$) for months preceding the start of the experiment. Standard errors are clustered at the consumer level. There are 40,708 credit cards with 368,162 observations.

Debt.—Figure 3, panel A, shows that our treatment does not significantly reduce credit card debt at or before the seventh cycle. Table 2 shows that the mean effect of the treatment on debt is an insignificant −0.5 percentage points, with a 95 percent confidence interval of −1.2 to 0.2 percentage points. As a robustness check, as part of our secondary analysis, we look at debt in pounds in Figure 3, panel B, and find precise zero effects. Table 3 shows that after seven cycles the treatment insignificantly changes debt by £4, with a 95 percent confidence interval of −£30 to £38. Examining the full distribution of debt reveals little discernible difference between these two measures (see Supplemental Appendix Figure A7 for details). Given that there is no significant reduction in debt, it is unsurprising that we also find no significant reduction in borrowing costs by our primary measure in Table 2 or other measures in Table 3. Tables 2 and 3 show that we do not find a significant effect of the nudge on reducing the portfolio of credit card debt held, when measured in percentages or pounds.

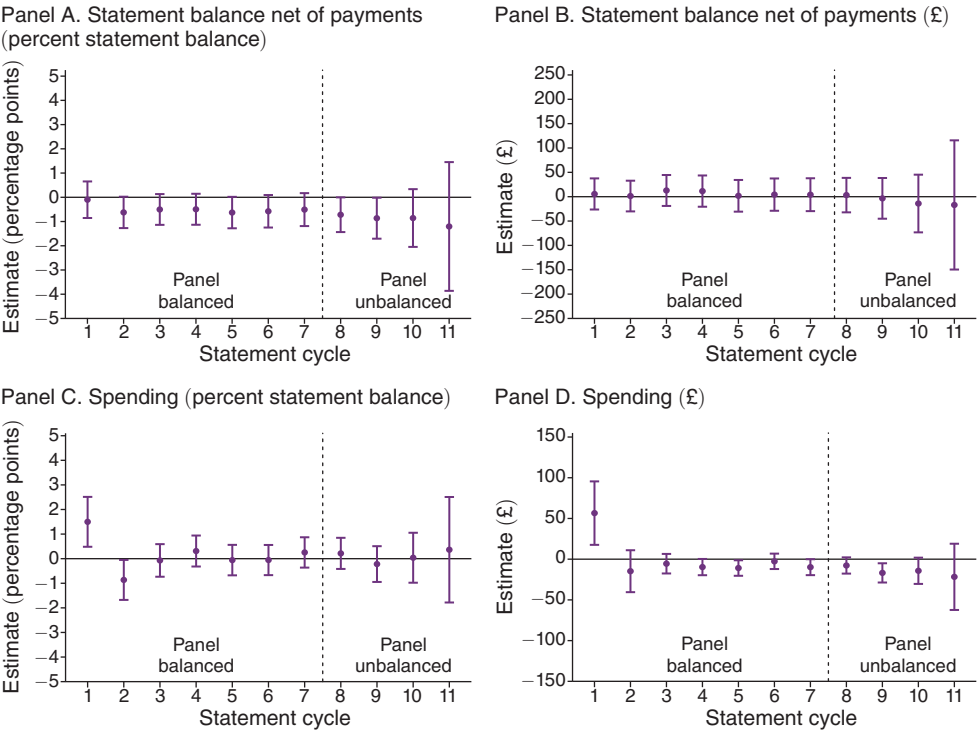


FIGURE 3. AVERAGE TREATMENT EFFECTS ON CREDIT CARD DEBT AND SPENDING, ACROSS 1–11 STATEMENT CYCLES

Notes: Treatment effects from coefficients (δ_t) on interaction terms between the treatment indicator and statement cycle indicators in the OLS regression specified in equation (1). Regression outcome in panel A is statement balance net of payments (percent statement balance), in panel B it is statement balance net of payments (£), in panel C it is spending (percent statement balance), and in panel D it is spending (£). Regressions also include statement-cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card cur-chases rate, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination, except for the variables constructed from consumer credit reporting data (credit score, DTI ratio, and any mortgage debt), which are from the month preceding card origination. Error bars are 95 percent confidence intervals. Standard errors are clustered at the consumer level. There are 40,708 credit cards with 368,162 observations.

As the cycle-by-cycle estimates on our primary measure of credit card debt are stable over time but persistently, slightly, but statistically insignificantly, below zero, we check the robustness of this result by pooling across all cycles to provide more statistical power (equation (2)). The average effect on debt on the target card is, at most, a 1.1 percentage points reduction, based on the maximum of the 95 percent confidence interval, and is insignificantly different from zero at our 0.5 percent threshold. Even with this pooling, there is no statistically significant effect on credit card debt across the portfolio of cards held, at most a 0.79 percentage point reduction, also based on the maximum of the 95 percent confidence interval. See Supplemental Appendix Table A5 for details.

Full Payment.—We find no effect of the treatment on increasing the probability of full payment on the target card. Table 2 shows that the average effect after seven

TABLE 3—AVERAGE TREATMENT EFFECTS FOR SECONDARY OUTCOMES AFTER SEVEN STATEMENT CYCLES

Outcome	Estimate (SE)	95% C.I.	<i>p</i> -value	Control mean
Statement balance (£)	−0.33 (17.24)	[−34.11, 33.46]	0.9848	2,164.49
Statement balance net of payments (£)	4.11 (17.22)	[−29.64, 37.85]	0.8115	1,962.52
Costs (£)	0.11 (0.18)	[−0.24, 0.46]	0.5312	7.97
Cumulative costs (£)	1.39 (0.83)	[−0.23, 3.01]	0.0924	76.02
Spending (£)	−9.84 (4.99)	[−19.61, −0.07]	0.0485	193.24
Cumulative spending (£)	−7.23 (20.95)	[−48.29, 33.83]	0.7300	3,186.19
Credit card portfolio statement balances (£)	23.65 (31.15)	[−37.42, 84.71]	0.4479	3,916.96
Credit card portfolio balances net of payments (£)	12.06 (30.92)	[−48.55, 72.66]	0.6966	3,431.69
Any autopay payment	0.0019 (0.0046)	[−0.0071, 0.0108]	0.6839	0.6171
Any manual payment	0.0064 (0.0045)	[−0.0024, 0.0151]	0.1537	0.3166
Total payments (£)	−4.44 (5.07)	[−14.38, 5.51]	0.3820	201.98
Automatic payments (£)	−0.6123 (2.1195)	[−4.7665, 3.5419]	0.7727	86.9490
Manual payments (£)	−3.90 (4.79)	[−13.29, 5.48]	0.4152	116.38
Total payments (% statement balance)	0.0060 (0.0032)	[−0.0002, 0.0123]	0.0579	0.2271

(continued)

cycles is 0.4 percentage points, and this is not significantly different from zero, with a 95 percent confidence interval of -0.3 to 1.1 . This finding is robust to pooling across all cycles, where the 95 percent confidence interval is -0.2 to 1.0 percentage points, with a point estimate of 0.4 percentage points (see Supplemental Appendix A5). This null result is also robust to measuring the effects on the cumulative number of full payments, as shown in Supplemental Appendix Table A4.

Spending.—One potential explanation for the nudge’s ineffectiveness in reducing debt would be if consumers responded by spending more. Our results rule out this explanation. Figure 3, panel C, and Table 3 show that we find null effects over time for new spending (percent statement balances), with a 95 percent confidence interval of -0.4 to 0.9 percentage points after seven cycles. Table 3 also shows that there is a null effect on statement balances (i.e., before payments, which reflects the combination of revolving debt and new spending). Figure 3, panel D, shows that there is a slight decrease in spending, measured in pounds of £9.84 (95 percent confidence interval $-£19.61, -£0.07$), however, this is not significant at our 0.5 percent threshold, and is insignificant when we study cumulative spending, shown in Table 3.

TABLE 3—AVERAGE TREATMENT EFFECTS FOR SECONDARY OUTCOMES AFTER SEVEN STATEMENT CYCLES
(continued)

Outcome	Estimate (SE)	95% C.I.	p-value	Control mean
Cumulative automatic payments (£)	27.30 (10.35)	[7.01,47.59]	0.0084	573.79
Cumulative manual payments (£)	−18.87 (13.97)	[−46.25,8.50]	0.1766	711.97
Cumulative payments (% cumulative spending)	0.0202 (0.0051)	[0.0101,0.0303]	0.0001	0.4728
Cumulative automatic payments (% cumulative spending)	0.0256 (0.0048)	[0.0162,0.0350]	0.0000	0.2135
Cumulative manual payments (% cumulative spending)	−0.0046 (0.0033)	[−0.0111,0.0019]	0.1662	0.2644
Cumulative payments (% credit limit)	0.0130 (0.0063)	[0.0006,0.0254]	0.0398	0.5046
Cumulative automatic payments (% credit limit)	0.0240 (0.0038)	[0.0165,0.0315]	0.0000	0.1864
Cumulative manual payments (% credit limit)	−0.0101 (0.0057)	[−0.0212,0.0011]	0.0760	0.3249

Notes: The table shows average treatment effects after seven statement cycles. Each row of the table shows estimates from separate regressions with different outcomes. Estimated treatment effects are from the coefficient (δ_7) on interaction terms between the treatment indicator and the seventh statement-cycle indicator in the OLS regression specified in equation (1). Regressions also include: interactions between treatment indicator and other statement cycles, statement cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card purchases rate, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (credit score, DTI ratio and any mortgage debt), which are from the month preceding card origination. Standard errors are clustered at the consumer level. There are 40,708 credit cards with 368,162 observations.

Missed Payments.—In Table 2, we find an increase in the probability of missed payments on the target card of 0.38 percentage points, with a 95 percent confidence interval of 0.02 to 0.75 percentage points, that is statistically significant at the 5 percent level but not at our 0.5 percent threshold. Although this increase is not statistically significant at our 0.5 percent significance threshold when examining a cycle, it is clearly significant when conducting a joint significance test pooling data across all cycles, while still clustering at the consumer level. We find that the nudge increases the probability of missed payments by 0.4 percentage points with a 95 percent confidence interval of 0.19 to 0.62. The effect on missed payments is solely that of being one payment behind. The nudge does not lead to consumers being in more severe arrears, which the industry defines as being at least two or three payments behind. These are all null results even when pooling observations across cycles to increase power to account for the low incidence of such severe arrears. Only more severe arrears are reported on their credit report (that is, missing a payment by one day would not be reported, but by 31 days would be reported). This explains why we do not observe increases in our Table 2 primary outcome of missed payments measured in credit reports. Given that there is no difference in severe arrears on the card in the experiment, and also no difference in severe arrears across the portfolio of

cards in credit reports, we infer that severe arrears on other cards are unaffected. See Supplemental Appendix Table A6 for estimates using different measures of arrears.

Other Portfolio Primary Outcomes.—Table 2 shows there are null effects on average treatment effects across our other consumer credit reporting outcomes: the probability of paying in full, the probability of missing payments, and outstanding debt when aggregating across the portfolio of credit cards held. There is no evidence that the treatment affects other cards held by a consumer. The results on primary outcomes are persistent over time (see Supplemental Appendix Figures A4 and A5).

C. Heterogeneous Effects

We examine heterogeneity in our results by covariates that are constructed from credit reporting data from the month preceding card origination or credit card data at origination. We estimate equation (1) separately for two quantiles of each variable, pooling data across cycles to increase power.¹² Consistent with our previous analysis, we find significant effects of the nudge to reduce Autopay Min enrollment, and also to reduce the fraction of consumers only paying exactly the minimum, across all heterogeneous groups examined (see Supplemental Appendix Table A10 for details).

Our heterogeneity analysis does not robustly show that debt is reduced by the nudge for a subgroup of consumers. We find suggestive evidence that the nudge reduces debt, measured as a percentage of the statement balance, for consumers with lower credit scores, lower unsecured debt-to-income (DTI) ratios, younger ages, lower credit limits, fewer credit cards in the portfolio with debt, lower credit card portfolio balances net of payments, and men, but no clear effects by income. None of these subgroups shows significant effects at the 5 percent significance level when debt is measured in pounds, however, and we also suggest that the reader applies a stricter threshold given concerns regarding multiple hypothesis testing. The largest heterogeneous effect is for low DTI consumers, where the nudge reduces debt, as a percent of statement balance, by -1.4 percentage points (standard error 0.4); however, this result is not robust, as the nudge insignificantly increases debt when measured in pounds by £2.74 (standard error £16.81). See Supplemental Appendix Table A11 for details.

IV. Mechanisms

A. Factors Explaining Nudge Ineffectiveness

Our null average treatment effects on debt are surprising, given the nudge has seemingly large changes in autopay enrollment and makes consumers less likely to only pay the minimum payment. Why does the treatment not, on average, reduce debt if one in five more consumers are enrolled in Autopay Fix, and it does not cause

¹² Analysis not in this version of the paper also found consistent results when not pooling data and instead focusing on the seventh statement cycle, and also when studying quartiles.

them to spend more? If the only effects of the nudge are compositional—changing autopay enrollment but *assuming no other changes*—then the effects on autopay enrollment may have been expected to lead to an effect of reducing debt by approximately 4.5 percent.¹³ Indeed, the fact that the second lender withdrew after only observing effects on enrollment is evidence that our null effects on the more distal debt outcome are unexpected. We examine how three consumer responses on the target card make the nudge ineffective at reducing debt.

Autopay Fix Amounts “Too Low”.—Cardholders often respond to the nudge by setting an Autopay Fix that is “too low,” binding at or just above the minimum. The purple coefficients in Figure 2, panel A, show that although the treatment causes an increase of 16.7 percentage points in Autopay Fix enrollment by statement seven, the pink coefficients show how the effect on enrollment with Autopay Fix *exceeding* the minimum payment amount due is still large, but is half the size: 8.6 percentage points, which is a 34 percent increase on the control group mean. See Table 1 for estimates. As credit card balances accumulate over the first few months of card ownership, the minimum amount due rises, causing the minimum payment amount to exceed many of the fixed payments. After seven cycles, the proportion of consumers in the treatment group with an Autopay Fix that exceeds the minimum payment amount is 66 percent, noticeably lower than 78 percent in the second cycle. See Supplemental Appendix Figure A9 for details.

Examining the distribution of Autopay Fix amounts chosen by the treatment group, they are often “low,” that is, small amounts in excess of the minimum. Pooling across all seven cycles, we find that for 48 percent of Autopay Fix enrollees in the treatment group, the cumulative Autopay Fix amount is £100 or less in excess of the minimum. The cumulative Autopay Fix amount is only more than £500, for 13 percent, where the control mean is £1,277. We interpret that the additional payments from Autopay Fix over the minimum are typically “low” in absolute levels; however, they are large increases relative to the extremely low minimum payment due, which averages £46 per month (£320 cumulative across cycles one to seven). This is consistent with studying excess payments: the value of payments (the sum of autopay and manual payments) in excess of the minimum, as a percentage of statement balances (also in excess of the minimum), measured after seven cycles. Excess payments for the treatment group, relative to the control group, are more likely to be 1 percentage point or more (56.9 percent in control and 60.2 percent in treatment); however, this difference quickly tends toward 0 as we move up the excess payments’ distribution: 2 percentage points or more is 50.8 percent in control and 52.9 percent in treatment, 3 percentage points or more is 46.9 percent in control and 48.3 percent in treatment, 5 percentage points or more is 41.2 percent in control and 42.0 percent in treatment, and 10 percentage points or more is 35.6 percent in control and 35.7 percent in treatment. The full distributions of Autopay Fix amounts and excess payments are shown, respectively, in Supplemental Appendix Figures A8 and A6. Consumers only selecting low Autopay Fix amounts, and making low payments in

¹³ Calculated using the mean debt net of payments in cycle seven for cardholders in the control group for each autopay enrollment type and weighting these by the treatment group’s autopay enrollments’ shares.

excess of the minimum when the Autopay Min option is shrouded, appears consistent with consumers only wanting to make low automatic payments, possibly due to uncertainty in how much liquid cash they will have to smooth their payments.

Decreasing Enrollment in Any Autopay.—The nudge causes a 4.3 percentage point or 5.6 percent significant decline in enrollment in any type of autopay, as shown in Table 1. This decline in enrollment explains an unintended slight average increase in the probability of missed payments, also shown in Table 1. If enrolled in autopay, a consumer will only miss a payment if they have insufficient funds in their checking account, whereas consumers not enrolled may easily forget to make a payment. Our earlier finding, that the effect is only on missing one payment, and not for severe arrears, repeatedly missing payments, indicates that not being enrolled in any autopay means consumers forget to make a payment, which has a temporary impact, most notably incurring a late payment fee and not reducing debt, rather than causing a debt spiral or severe distress. Although reduced enrollment in autopay is not an intended effect of the nudge, it does not increase consumer indebtedness. These results are consistent with consumers being more attentive to their debt if they do not enroll in autopay (Gathergood et al. 2021; Sakaguchi et al. 2022).¹⁴ This is different from other domains where reduced enrollment may be a worse economic outcome. For example, if a nudge reduces 401(k) enrollment, then consumers can miss out on “free money” from employer-matched contributions and under-save for retirement.

Our explanation is also consistent with the analysis calculating “Lee bounds” (Lee 2009), as used in Levy (2021). This methodology provides bounds on the treatment effects after accounting for selection in outcomes that occur due to the nudge reducing autopay enrollment. These bounds require the monotonicity assumption that the nudge only makes consumers less likely to enroll in any autopay. Pooling data across all cycles, we estimate Lee bounds that the treatment causes a -12.27 to -6.42 percentage point change in the probability of only making a minimum payment. Lee bounds also show that the nudge causes a -0.01 to 0.41 percentage point change in any missed payments and a 0.04 to 0.09 percentage point change in missing two or more payments. The Lee bounds also show that effects are insignificant from zero for debt, statement balances, costs, and spending, as shown in Supplemental Appendix Table A9.

Manual Payments Substitution.—Cardholders can make manual payments instead of, or in addition to, automatic payments. We find substitution between the two as another offsetting effect. Figure 4 and Table 3 show that there is a positive and significant treatment effect of increasing automatic payments; however, the effect on overall payments is smaller due to a negative, but statistically insignificant, effect on manual payment. We find that the treatment causes consumers to be 1.3 percentage points more likely to make both an automatic and manual payment in the same cycle, despite reduced autopay enrollment. More details are in Supplemental Appendix Table A4.

¹⁴This is consistent with another domain; Sexton (2015) argues that enrollment into autopay (full) for utility bills, reduces price salience and results in “overconsumption” of electricity.

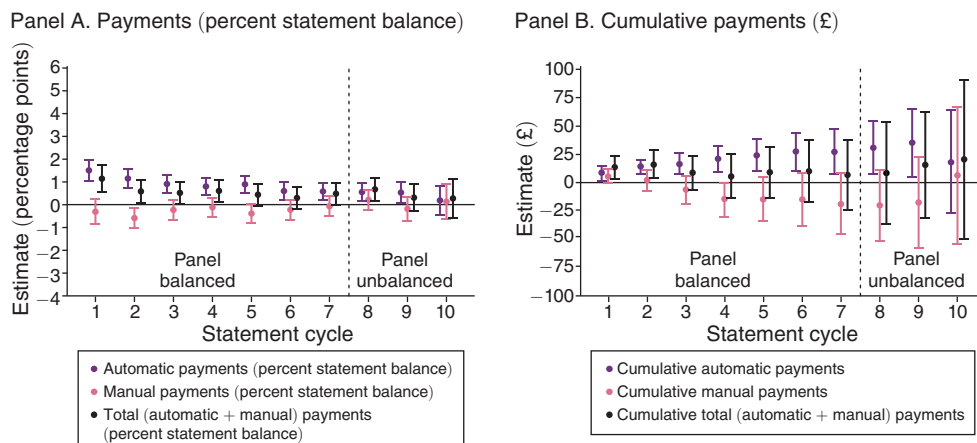


FIGURE 4. AVERAGE TREATMENT EFFECTS ON AUTOMATIC, MANUAL, AND TOTAL (AUTOMATIC + MANUAL) PAYMENTS, ACROSS 1–10 STATEMENT CYCLES

Notes: Treatment effects from coefficients (δ_T) on interaction terms between the treatment indicator and statement cycle indicators in the OLS regression specified in equation (1). The regression outcome in panel A is payments (percent statement balance), and the outcome in panel B is cumulative payments (£). Regressions also include statement-cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgaged, log credit card credit limit, credit card purchases rate, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (credit score, DTI ratio and any mortgage debt), which are from the month preceding card origination. Error bars are 95 percent confidence intervals. Standard errors are clustered at the consumer level. Cycle 11 is excluded from this figure as, due to few cards being observed in this cycle, the confidence intervals on panel B are extremely large such that the estimates are uninformative. There are 40,708 credit cards with 368,162 observations.

Manual payments are infrequent, but large. Only 8.5 percent of those enrolled in any autopay option in the control group also made a manual payment in the seventh cycle. However, manual payments account for 45 percent of the total cumulative value of payments made across cycles one to seven by those in the control group enrolled in autopay at cycle seven (and 54 percent for those enrolled in Autopay Fix or Min). In months when manual payments are made by those enrolled in autopay in the control group, the mean manual payment amount is £377, with a median value of £105. By contrast, automatic payments in these months average £105 with a median of £55. We interpret this evidence as showing that consumers appear to use autopay as insurance against forgetting to make a payment (in line with Fuentealba et al. 2021; Gathergood et al. 2021; Sakaguchi et al. 2022), as opposed to paying down debt. Survey responses presented in our earlier working papers (Adams et al. 2018a, b) align with this explanation.¹⁵

¹⁵ In Adams et al. (2018a, b), the most common reasons survey respondents enrolled in autopay provide for using autopay is to prevent incurring a late fee or to prevent a negative credit score impact, while the most common reason respondents not enrolled in autopay provide is that they prefer the control of manually adjusting payments each month.

Comparing automatic and manual payments conflates two effects: a change in autopay enrollment composition and a change in autopay amount. Conditional on being enrolled in autopay, one would expect automatic payments to be higher in the treatment than the control, since Autopay Fix is greater than or equal to Autopay Min. Automatic payments will be lower in the treatment group because fewer consumers enroll in autopay than in the control group. Similarly, we may expect manual payments to be higher in the treatment group; however, this is ambiguous, as cardholders may forget to make any payments rather than substitute automatic for manual payments. We unravel this by decomposing equation (1) by whether the consumer is enrolled in any autopay (that is, Autopay Min, Fix, or Full) in cycle seven ($AUTOPAY_{i,7}$), shown in equation (3). This is a decomposition by an endogenous variable, and therefore, the estimates are not causal and may be biased.

$$(3) \quad Y_{i,t} = \alpha + \sum_{\tau=1}^T \delta_{\tau}(TREATMENT_i \times CYCLE_{\tau}) + \mathbf{X}'_i \beta + \gamma_{m(i,t)} + \gamma_t + \varepsilon_{i,t}$$

if $AUTOPAY_{i,7} = g, g \in \{0,1\}$.

We examine the cumulative value of payments, in total and split by automatic and manual payments, by the seventh cycle in these subgroups in Figure 5. Panel A of Figure 5 shows evidence of substitution among consumers enrolled in autopay: The average change in automatic payments is an increase of £62, average manual payments decrease by £57, and the average overall payments are effectively unchanged, with an increase of only £2. If all the increased automatic payments had passed through, without offsetting manual payments, average debt would have been reduced by approximately 2.9 percent. Panel B of Figure 5 shows zero estimates on automatic, manual, and total payments for those not enrolled in autopay. This indicates the treatment's main effect on this group is likely shifting this group's size rather than changing its payment amounts differentially to what one would expect from a cardholder in the control group who is not enrolled in autopay. These results are robust to alternative measurement approaches shown in Supplemental Appendix Figure A11. Our manual payment substitution result is also robust to calculations using Lee bounds. The nudge causes an increase in automatic payments with bounds of −£2.03 to £42.10, or, as a percentage of statement balances, is 0.46 to 5.71 percentage points. Manual payments attenuate this effect, leading to bounds on total payments of −£19.01 to £84.66, or, as a percent of statement balances, is −0.08 to 4.35 percentage points; see Supplemental Appendix Table A9 for details. This substitution of higher automatic payments for lower manual payments by cardholders enrolled in autopay shows that these consumers are less inert in their payment choices than they initially appeared. The most natural explanation for such substitution is that these consumers often lack liquid cash to pay more on their credit card. If so, once a consumer has made slightly higher automatic payments, there is less cash available for manual payments.

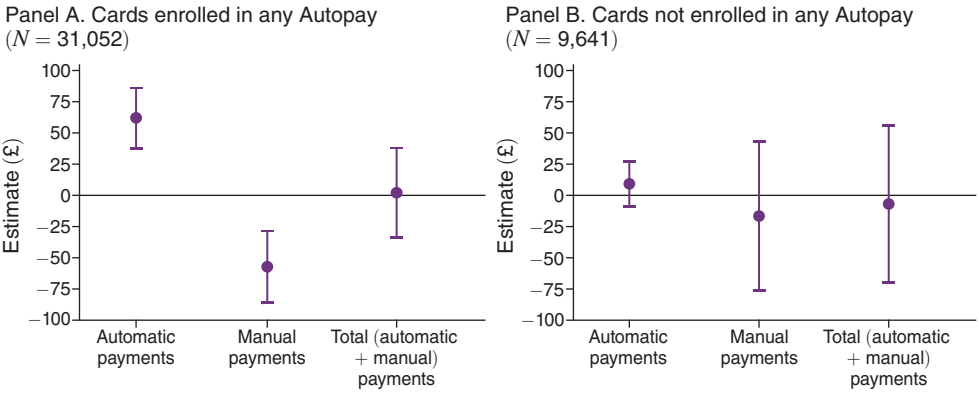


FIGURE 5. NON-CAUSAL DECOMPOSITION OF ESTIMATES ON CUMULATIVE PAYMENTS, BY ANY AUTOPAY ENROLLMENT AFTER SEVEN STATEMENT CYCLES

Notes: Each panel shows outcomes from two separate regressions where outcomes are: cumulative automatic payments, cumulative manual payments, and cumulative total (automatic + manual) payments. Regressions also include: interactions between the treatment indicator and other statement cycles, statement cycle fixed effects, year-month fixed effects, and the following controls: gender, age, age squared, log estimated income, credit score, unsecured debt-to-income (DTI) ratio, any mortgage debt, log credit card credit limit, credit card purchases rate, subprime credit card, any credit card promotional rate, any credit card balance transfer, credit card open date, credit card statement day, any credit card secondary cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (credit score, DTI ratio, and mortgage debt), which are from the month preceding card origination. Error bars are 95 percent confidence intervals. Standard errors are clustered at the consumer level. Panels A and B show noncausal estimates (δ_7) from the OLS regression specified in equation (3), which has the same specification except that panel A restricts to the subsample of 31,052 credit cards that are enrolled in any autopay at statement-cycle seven, and panel B restricts to the subsample of 9,641 credit cards that are not enrolled in any autopay at statement cycle seven.

B. Liquid Cash

Having documented the effects of the nudge and investigated the factors that explain our null result, we wanted to understand *why* consumers are not paying more on their credit cards. Doing so can help to understand whether other interventions may be more effective than the one nudge we test. The most natural potential explanation is that many consumers in our study have limited liquid cash balances available, which prevents or discourages them from reducing their credit card debt. While one may term limited liquid cash balances as liquidity constraints, we caveat that limited liquid cash balances are an observable financial outcome that may arise for many reasons, including financial illiteracy (e.g., Lusardi and Tufano 2015) and behavioral factors, such as naïve present bias, leading to impulsive overconsumption (e.g., Heidhues and Köszegi 2015), which are well-documented among credit cardholders (e.g., Meier and Sprenger 2010; Kuchler and Pagel 2021).

In this section, we conduct an analysis using linked bank account data for three purposes. First, we evaluate the prevalence of low liquid cash balances among the selected sample where the linked data are observed. If few consumers have liquid cash available, it means that few are likely to be able to pay more of their credit

card debt, even if nudged. Second, we study whether, among this selected sample, these liquid cash balances explain credit card behaviors seven months later. If there is a relationship between measures of liquid cash balances and subsequent credit card debt, then this means these measures are informative for understanding the credit card market. Third, we examine whether the nudge heterogeneously affects consumers by their liquid cash balances, again for this selected sample. We might expect the nudge to be effective in reducing the debt of consumers who are not liquidity-constrained and so could afford to pay more when the minimum is shrouded. All three purposes represent an advance on prior research on credit card anchoring, where liquid cash balances and also autopay, are unobserved (Keys and Wang 2019; Medina 2021). Studies of co-holding (e.g., Vihriälä 2025; Batista, Mao and Sussman 2025; Gathergood and Olafsson 2024) have a different focus, studying the *simultaneous* holding of low-yield liquid cash while also carrying high-interest credit card or overdraft debt, whereas we study whether the *dynamics* of liquid cash *before* card opening are informative for understanding credit card debt accumulated *months later*.

Measuring Liquid Cash.—We construct measures of liquid cash before card opening from our linked bank account data. We calculate liquid cash as the end-of-day balance aggregated across all observed checking and non-checking, instantly accessible cash savings accounts. In the United Kingdom, checking accounts often have an overdraft line of credit facility, so liquid cash measures can have negative balances. Based on observed socioeconomic characteristics, we expect the selected sample with linked bank account data to be less liquidity constrained than other consumers in our field experiment. See Supplemental Appendix E for more details. We construct two measures to capture the dynamics of liquid cash balances.¹⁶ For each measure, we show, in Figure 6, its CDF in panels A and B, and its relationship with credit card payments seven cycles later in panels C and D. Our two measures are:

- The *Liquid Cash Balances* of a consumer on the day before credit card opening. This traditional static measure reflects a consumer's liquid cash balances at one point-in-time. The different colors in panel A of Figure 6 show that the overall distribution of this variable is stable when calculated at different points-in-time, 30 to 150 days before card opening; however, it may vary for individual consumers over time. We also observe that this distribution has very fat tails, and so the mean is not well-estimated.
- The *Minimum Liquid Cash Balances* over the last 90 days before card opening. This is a new dynamic measure that we have introduced to the literature to overcome the weakness that point-in-time measures do not reflect how some consumers may only temporarily hold high balances, potentially due to a mismatch in the timing of their incomes and expenditures. The different colors in panel B of Figure 6 show that the S-shaped distribution of this variable steepens around

¹⁶ Supplemental Appendix C contains additional results for a third measure based on the number of days liquid cash balances are below £100 in the 30 days before card opening.

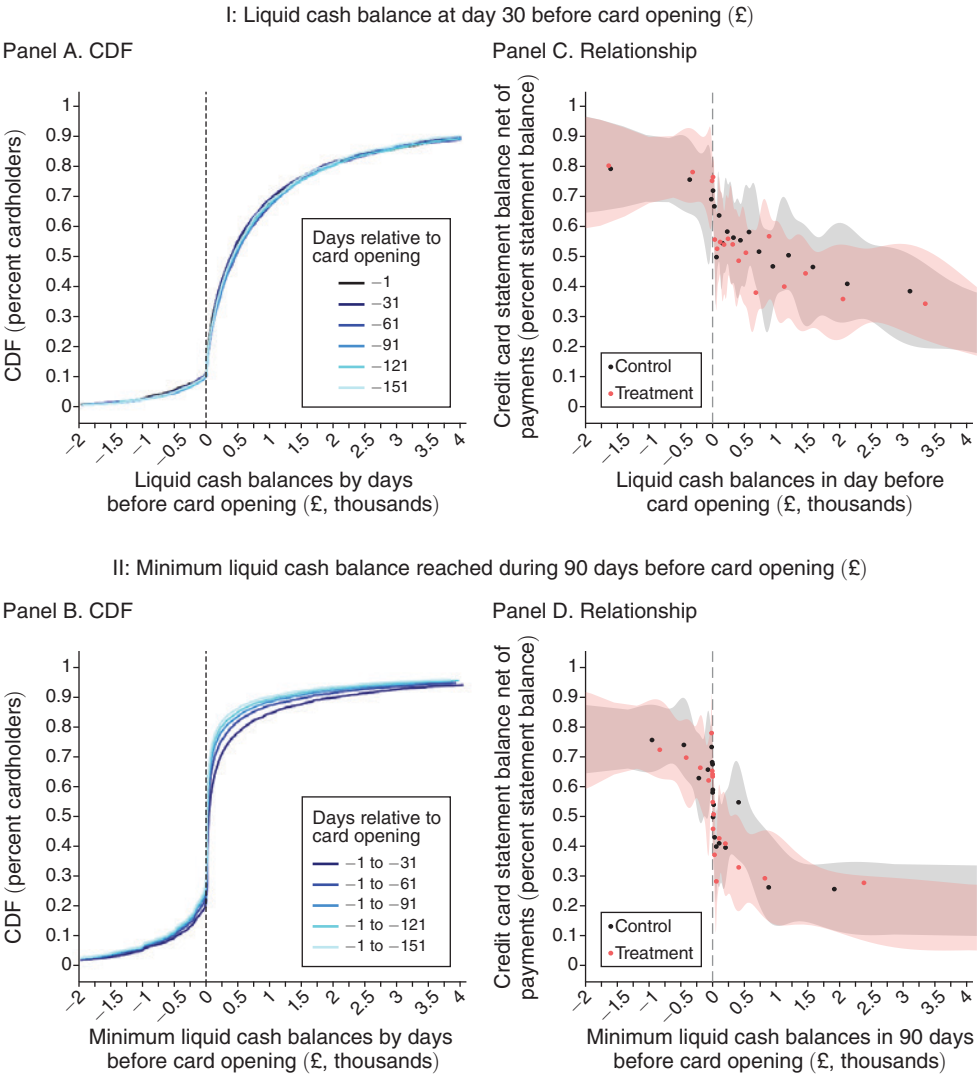


FIGURE 6. CDFS OF LIQUID CASH BALANCES MEASURED BEFORE CARD OPENING (PANELS A AND B) AND THEIR NONPARAMETRIC RELATIONSHIPS WITH CREDIT CARD DEBT (STATEMENT BALANCE NET OF PAYMENTS AS A PERCENT OF STATEMENT BALANCE) AT STATEMENT-CYCLE SEVEN, BY TREATMENT GROUP (PANELS C AND D)

Notes: $N = 3,753$ consumers. Liquid cash balances are measured before credit card opening. Panels A and B are CDFS. Panels C and D are binned scatterplots by quantiles of the distribution (Cattaneo et al. 2024) where error bands are 95 percent confidence intervals. The x -axes of all panels are censored to ease presentation given a fat tail to the distribution of these variables.

zero when calculated over longer time windows, up to 150 days before card opening, and flattens over shorter time windows, down to 30 days before card opening. We also note that this distribution has very fat tails, and so the mean is not well-estimated. This is a new measure; prior literature has only studied different moments: the point-in-time, mean, or median balances.

Liquid Cash before Card Opening.—Examining our first static measure initially suggests that consumers generally have liquid cash available before opening the card. Panel A of Figure 6 shows the cumulative distribution of our liquid cash balance measure and indicates that consumers commonly have funds available: Only 13 percent of consumers have liquidity of a zero or negative liquid cash balance, the median has £370 available, and 30 percent appear relatively unconstrained with over £1,000. This measure shows a clear kink with liquid cash balance above zero being much more likely than those below. This kink may reflect the discontinuous increase in costs from becoming overdrawn on checking accounts and a precautionary rationale to keep a small amount of buffer stock savings.

It is only once we examine our second measure, minimum liquid cash balances, that captures the dynamics of liquid cash balances that we reveal that consumers frequently lack liquid cash. Figure 6, panel B shows the cumulative distribution of this measure. There is substantial bunching of consumers just managing to keep positive, but small, liquid cash balances. Half of consumers have effectively zero minimum liquid cash balances in the 90 days before card opening (median £5). Many consumers' minimum liquid balances are only for small amounts. For example, the 75 percentile is only £142, 84 percent have £500 or less, and 89 percent have £1,000 or less. These reveals more consumers have limited liquid cash than the point-in-time liquid cash balance measure indicate, for example the 75 percentile of the point-in-time measure is £1,311. This lack of liquid cash among many consumers in our experiment helps to explain why the nudge was ineffective. See Supplemental Appendix Table C1 for detailed summary statistics of both measures. These liquid cash measures have weak correlations with covariates typically observed by researchers without access to linked bank account data, see Supplemental Appendix Table C2 for details.

Relationship between Liquid Cash and Credit Card Debt.—We show that these measures of liquid cash before card opening explain credit card debt held seven cycles later. When examining credit card debt, we note that this is a function of both spending and payment decisions. Panels C and D of Figure 6 plot binned scatterplots, using the methodology of Cattaneo et al. (2024), as a clean way to visualize nonlinear relationships between liquid cash measures on the x -axes and credit card debt on the y -axes. Panel C of Figure 6 shows that consumers who had small positive liquid cash balances before card opening, our first measure, held less credit card debt, on average, seven cycles later than those with zero or small negative liquid cash balances. Panel D of Figure 6 shows that consumers with positive minimum liquid balances before card opening, our second measure, discontinuously hold approximately 20 percentage points less credit card debt, as a percentage of statement balance, seven cycles later than those with zero or small negative minimum liquid balances.

Given the bimodal distribution of payments, we also examine the other moments: payments at the minimum, full, and less than minimum amounts. The discontinuity in average payments is driven by discontinuous increases in the probability of paying in full and decreases in the probability of missing a payment. Approximately 70 percent of the small subset of consumers with high minimum liquid cash balances (e.g., £1,000+) pay in full, and these are largely enrolled in Autopay Full, suggesting that

there is limited room for the nudge to increase payments among this group. The relationship with autopay enrollment choices is less clear, except for a discontinuous increase in Autopay Full enrollment. Paying only the minimum becomes less likely among less liquidity-constrained consumers; however, there is a less clear discontinuity around zero. See Supplemental Appendix Figure C2 for details. Our dynamic measures appear to be important to understanding credit card usage, and, as shown in Figure 6, there are nonlinearities in the relationship between liquid cash and credit card debt that simple correlations would not capture. Overall, this means that the high prevalence of low liquid cash among consumers in our experiment, which we show predicts low credit card debt as a percent of statement balances, potentially helps to explain the nudge's ineffectiveness. Consumers appear to be making "low" credit card payments and offsetting the nudge to not reduce their debt due to frequently holding limited liquid cash balances. Such limited liquidity may also mean other policies intended to change consumer choices around repayment may ultimately fail to reduce credit card debt.

Heterogeneous Effects of Nudge by Liquid Cash.—As shown in the previous sections, very few consumers consistently have nontrivial amounts of liquid cash before card opening. We examine whether the nudge is effective for the small minority of consumers with high liquid cash before card opening. We pool data across all cycles to increase power and show sensitivity to alternative thresholds when defining consumers as holding high liquid cash. Across all heterogeneous groups of high and low liquid cash, we find that the nudge significantly reduces the probability of enrolling in any autopay and the probability of consumers only paying exactly the minimum. Details are in Supplemental Appendix Table C3. There is very weak evidence that the nudge causes some high liquid cash consumers to reduce their debt; however, this result is not at all robust. There are declines for consumers with high liquid cash as measured by the liquid cash balance of £501+ (−3.42 percentage points, 95 percent confidence interval −6.35 to −0.49 percentage points) and £1,001+ (−4.48 percentage points, 95 percent confidence interval −7.95 to −1.00 percentage points), both significant at the 5 percent level, and insignificant declines for other groups of consumers (< £1, £1+, < £501, and < £1,001). By our second measure, there are very few high-liquidity consumers and, therefore, we do not have the power to conclude whether there is an imprecisely estimated significant effect or a null result; for example, the 95 percent confidence interval for the £1,001+ group is −6.57 to 3.31. No heterogeneous group of either measure shows a decrease in credit card debt, measured in pounds, that is close to being statistically significant, with the smallest *p*-value at 0.29. As in our earlier heterogeneous analysis, we also note that the reader may want to apply a stricter threshold than 0.5 percent to evaluate significance, given concerns over multiple hypothesis testing. See Supplemental Appendix Table C4 for details.

While we cannot precisely rule out whether the effects for the small group of high liquid cash consumers are zero or not, we can test whether effects for these consumers are significantly different from those estimated for low liquid cash consumers. *t*-tests show that this is the case; however, results are sensitive to whether the debt outcome is measured as a percent of the statement balance or in pounds

(£), the liquid cash measure used, and the threshold to classify consumers as holding high liquid cash. The two robust results across outcomes and time periods are when liquid cash balances are £1,001+ or when minimum liquid cash balances are £501+, the effects of the nudge are significantly stronger, reducing debt in comparison to consumers with lower liquid cash. When liquid cash balances are £1,001+, compared to < £1,001, there is a significant differential effect of the nudge on debt: −3.3 percentage points (95 percent confidence interval −3.4 to −3.2 percentage points) and −£38 (95 percent confidence interval −£42 to −£34). The effect of the nudge on debt after seven cycles is significantly stronger when minimum liquid cash balances are £501+, compared to < £501: −1.2 percentage points (95 percent confidence interval −1.3 to −1.1) and −£66 (95 percent confidence interval −£71 to −£62). See Supplemental Appendix Table C5 for more details. These results therefore help to understand why consumers in our experiment are less “nudge-able” than they first appeared from their autopay choices and inert minimum payment behavior. Most consumers lack liquid cash.

V. Conclusions

We expected that successfully nudging credit cardholders away from enrolling in autopay set to only pay exactly the minimum payment would reduce their debt. Our field experiment shows that this did not occur. We show how an active choice nudge is successful in significantly and persistently changing *proximate* outcomes: decreasing Autopay Min enrollment by 74 percent and decreasing the probability of only paying exactly the minimum by 23 percent after six months. However, the nudge is unsuccessful at significantly reducing the *distal* outcome of the amount of credit card debt accumulated after six months. This is explained by offsetting consumer responses consistent with consumers frequently holding low liquid cash balances. Our study is an example of the difficulty of changing distal consumer behaviors and highlights the need to evaluate policies, such as nudges, on how they impact *distal* rather than *proximate* outcomes.

REFERENCES

- Abadie, Alberto. 2020. “Statistical Nonsignificance in Empirical Economics.” *American Economic Review: Insights* 2 (2): 193–208.
- Adams, Paul, Benedict Guttman-Kenney, Lucy Hayes, Stefan Hunt, David Laibson and Neil Stewart. 2022a. *The Semblance of Success in Nudging Consumers to Pay Down Credit Card Debt*. AEA RCT Registry. <https://doi.org/10.1257/rct.9326-1.1>.
- Adams, Paul, Benedict Guttman-Kenney, Lucy Hayes, Stefan Hunt, David Laibson, and Neil Stewart. 2022b. “Do Nudges Reduce Borrowing and Consumer Confusion in the Credit Card Market?” *Economica* 89 (S1): S178–S199.
- Adams, Paul, Benedict Guttman-Kenney, Lucy Hayes, Stefan Hunt, David Laibson and Neil Stewart. 2018a. “The Conflict between Consumer Intentions, Beliefs and Actions to Pay Down Credit Card Debt.” Financial Conduct Authority Occasional Paper 44.
- Adams, Paul, Benedict Guttman-Kenney, Lucy Hayes, Stefan Hunt, David Laibson and Neil Stewart. 2018b. “The Semblance of Success in Nudging Consumers to Pay Down Credit Card Debt.” Financial Conduct Authority Occasional Paper 45.
- Agarwal, Sumit, Souphala Chomsisengphet, Neale Mahoney, and Johannes Stroebe. 2015. “Regulating Consumer Financial Products: Evidence from Credit Cards.” *Quarterly Journal of Economics* 130 (1): 111–64.

- Agarwal, Sumit, Souphala Chomsisengphet, Neale Mahoney, and Johannes Stroebe. 2017. "Do Banks Pass through Credit Expansions to Consumers Who Want to Borrow?" *Quarterly Journal of Economics* 133 (1): 129–90.
- Allcott, Hunt, Daniel Cohen, William Morrison, and Dmitry Taubinsky. 2024. "When Do "Nudges" Increase Welfare?" *American Economic Review* 115 (5): 1555–96.
- Athey, Susan, Raj Chetty, Guido W. Imbens, and Hyunseung Kang. Forthcoming. "The Surrogate Index: Combining Short-Term Proxies to Estimate Long-Term Treatment Effects More Rapidly and Precisely." *Review of Economic Studies*.
- Bartels, Daniel M., Nicholas Herzog, and Abigail B. Sussman. 2023. "Distinguishing between Anchors and Targets." https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4521038.
- Batista, Rafael, Ella Mao, and Abigail B. Sussman. 2025. "Keeping Cash and Revolving Debt: How Consumers' Preference for Spending on Debit versus Credit Influences their Decision to Co-Hold." https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5289830.
- Benjamin, Daniel J., James O. Berger, Magnus Johannesson, Brian A. Nosek, E.-J. Wagenmakers, Richard Berk, Kenneth A. Bollen, et al. 2018. "Redefine Statistical Significance." *Nature Human Behaviour* 2 (1): 6–10.
- Beshears, John, James J. Choi, David Laibson, and Brigitte C. Madrian. 2018. "Behavioral House-Hold Finance." In *Handbook of Behavioral Economics: Applications and Foundations I*, Vol. 1, edited by B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, 177–276. Elsevier.
- Beshears, John, and Harry Kosowsky. 2020. "Nudging: Progress to Date and Future Directions." *Organizational Behavior and Human Decision Processes* 161 (Supplement): 3–19.
- Carroll, Gabriel D., James J. Choi, David Laibson, Brigitte C. Madrian, and Andrew Metrick. 2009. "Optimal Defaults and Active Decisions." *Quarterly Journal of Economics* 124 (4): 1639–74.
- Cattaneo, Matias D., Richard K. Crump, Max H. Farrell, and Yingjie Feng. 2024. "On Binscatter." *American Economic Review* 114 (5): 1488–514.
- Chetty, Raj, John N. Friedman, Søren Leth-Petersen, Torben Heien Nielsen, and Tore Olsen. 2014. "Active vs. Passive Decisions and Crowd-Out in Retirement Savings Accounts: Evidence from Denmark." *Quarterly Journal of Economics* 129 (3): 1141–219.
- Choukhmane, Taha. Forthcoming. "Default Options and Retirement Saving Dynamics." *American Economic Review*.
- Consumer Financial Protection Bureau. 2021. *The Consumer Credit Card Market*. Bureau of Consumer Financial Protection. https://files.consumerfinance.gov/f/documents/cfpb_consumer-credit-card-market-report_2021.pdf.
- DellaVigna, Stefano, and Elizabeth Linos. 2022. "RCTs to Scale: Comprehensive Evidence from Two Nudge Units." *Econometrica* 90 (1): 81–116.
- Financial Conduct Authority. 2014. *Credit Card Market Study Terms of Reference*. Financial Conduct Authority. <https://www.fca.org.uk/publication/market-studies/ms14-6-1.pdf>.
- Financial Conduct Authority. 2016a. *Credit Card Market Study Annex 4: Behavioural Trials*. Financial Conduct Authority. <https://www.fca.org.uk/publication/market-studies/ms14-6-3-credit-card-market-study-final-findings-report-annex-4.pdf>.
- Financial Conduct Authority. 2016b. *Credit Card Market Study Final Findings Report*. Financial Conduct Authority. <https://www.fca.org.uk/publication/market-studies/ms14-6-3-credit-card-market-study-final-findings-report.pdf>.
- Fuentealba, Carlos L. Delgado, Jorge A. Muñoz Mendoza, Sandra M. Sepúlveda Yelpeo, Carmen L. Veloso Ramos, and Rodrigo A. Fuentes-Solís. 2021. "Household Debt, Automatic Bill Payments, and Inattention: Theory and Evidence." *Journal of Economic Psychology* 85: 102385.
- Gathergood, John, and Arna Olafsson. 2024. "The Coholding Puzzle: New Evidence from Transaction-Level Data." *Review of Financial Studies*. Forthcoming.
- Gathergood, John, Hiroaki Sakaguchi, Neil Stewart, and Jörg Weber. 2021. "How do Consumers Avoid Penalty Fees? Evidence from Credit Cards." *Management Science* 67 (4): 2562–78.
- Gibbs, Christina, Benedict Guttman-Kenney, Donghoon Lee, Scott Nelson, Wilbert van der Klaauw, and Jialan Wang. 2025. "Consumer Credit Reporting Data." *Journal of Economic Literature* 63 (2): 598–636.
- Gomes, Francisco, Michael Haliassos, and Tarun Ramadorai. 2021. "Household Finance." *Journal of Economic Literature* 59 (3): 919–1000.
- Gross, David B., and Nicholas S. Souleles. 2002. "Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data." *Quarterly Journal of Economics* 117 (1): 149–85.
- Guttman-Kenney, Benedict. 2024. "Essays on Household Finance." PhD diss., University of Chicago.

- Guttman-Kenney, Benedict, Paul Adams, Stefan Hunt, David Laibson, Neil Stewart, and Jesse Leary.** 2023. "The Semblance of Success in Nudging Consumers to Pay Down Credit Card Debt." NBER Working Paper 31926.
- Guttman-Kenney, Benedict, Paul Adams, Stefan Hunt, David Laibson, Neil Stewart, and Jesse Leary.** 2025. *Data and Code for "The Semblance of Success in Nudging Consumers to Pay Down Credit Card Debt."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E209827V1>.
- Guttman-Kenney, Benedict, Jesse Leary, and Neil Stewart.** 2018. "Weighing Anchor on Credit Card Debt." Financial Conduct Authority Occasional Paper No. 43.
- Guttman-Kenney, Benedict., and Andrés Shahidinejad.** 2025. "Unraveling Information Sharing in Consumer Credit Markets." https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4629496.
- Heidhues, Paul, and Botond Köszegi.** 2015. "On the Welfare Costs of Naiveté in the US Credit-Card Market." *Review of Industrial Organization* 47 (3): 341–54.
- Hershfield, Hal E., and Neal J. Roese.** 2015. "Dual Payoff Scenario Warnings on Credit Card Statements Elicit Suboptimal Payoff Decisions." *Journal of Consumer Psychology* 25 (1): 15–27.
- Karlan, Dean, and Jonathan Zinman.** 2012. "Borrow Less Tomorrow: Behavioral Approaches to Debt Reduction." https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2060548.
- Keller, Punam Anand, Bari Harlam, George Loewenstein, and Kevin G. Volpp.** 2011. "Enhanced Active Choice: A New Method to Motivate Behavior Change." *Journal of Consumer Psychology* 21 (4): 376–83.
- Keys, Benjamin J., and Jialan Wang.** 2019. "Minimum Payments and Debt Paydown in Consumer Credit Cards." *Journal of Financial Economics* 131 (3): 528–48.
- Kuchler, Theresa, and Michaela Pagel.** 2021. "Sticking to Your Plan: Empirical Evidence on the Role of Present Bias for Credit Card Paydown." *Journal of Financial Economics* 139 (2): 359–88.
- Laibson, David.** 2020. Nudges are Not Enough: The Case for Price-Based Paternalism. AEA/AFA Joint Luncheon.
- Lee, David S.** 2009. "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects." *Review of Economic Studies* 76 (3): 1071–102.
- Levy, Ro'ee.** 2021. "Social Media, News Consumption, and Polarization: Evidence from a Field Experiment." *American Economic Review* 111 (3): 831–70.
- List, John A., Matthias Rodemeier, Sutanuka Roy, and Gregory K. Sun.** 2023. "Judging Nudging: Understanding the Welfare Effects of Nudges Versus Taxes." NBER Working Paper 31152.
- Lusardi, Annamaria, and Peter Tufano.** 2015. "Debt Literacy, Financial Experiences, and Overindebtedness." *Journal of Pension Economics & Finance* 14 (4): 332–68.
- Madrian, Brigitte C., and Dennis F. Shea.** 2001. "The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior." *Quarterly Journal of Economics* 116 (4): 1149–87.
- Marbach, Moritz, and Dominik Hangartner.** 2020. "Profiling Compliers and Noncompliers for Instrumental-Variable Analysis." *Political Analysis* 28 (3): 435–44.
- Medina, Paolina C.** 2021. "Side Effects of Nudging: Evidence from a Randomized Intervention in the Credit Card Market." *Review of Financial Studies* 34 (5): 2580–607.
- Medina, Paolina C., and Jose L. Negrin.** 2022. "The Hidden Role of Contract Terms: The Case of Credit Card Minimum Payments in Mexico." *Management Science* 68 (5): 3856–77.
- Meier, Stephan, and Charles Sprenger.** 2010. "Present-Biased Preferences and Credit Card Borrowing." *American Economic Journal: Applied Economics* 2 (1): 193–210.
- Navarro-Martinez, Daniel, Linda Court Salisbury, Katherin N. Lemon, Neil Stewart, William J. Matthews, and Adam J. L. Harris.** 2011. "Minimum Required Payment and Supplemental Information Disclosure Effects on Consumer Debt Repayment Decisions." *Journal of Marketing Research* 48 (SPL): S60–S77.
- Saccardo, Silvia, Hengchen Dai, Maria Han, Naveen Raja, Sitaram Vangala, and Daniel Croymans.** 2025. "Scaling Nudges: Who Moves and How." https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3971192.
- Sakaguchi, Hiroaki, Neil Stewart, John Gathergood, Paul Adams, Benedict Guttman-Kenney, Lucy Hayes, and Stefan Hunt.** 2022. "Default Effects of Credit Card Minimum Payments." *Journal of Marketing Research* 59 (4): 775–96.
- Schwartz, Daniel.** 2024. "The Rise of a Nudge: Field Experiment and Machine Learning on Minimum and Full Credit Card Payments." Unpublished.
- Seira, Enrique, Alan Elizondo, and Eduardo Laguna-Müggenburg.** 2017. "Are Information Disclosures Effective? Evidence from the Credit Card Market." *American Economic Journal: Economic Policy* 9 (1): 277–307.

- Sexton, Steven.** 2015. "Automatic Bill Payment and Salience Effects: Evidence from Electricity Consumption." *Review of Economics and Statistics* 97 (2): 229–41.
- Stewart, Neil.** 2009. "The Cost of Anchoring on Credit-Card Minimum Repayments." *Psychological Science* 20 (1): 39–41.
- Thaler, Richard H.** 2017. "Much Ado about Nudging." *Behavioral Public Policy Blog*. <https://bppblog.com/2017/06/02/much-ado-about-nudging/>.
- Thaler, Richard H., and Shlomo Benartzi.** 2004. "Save More Tomorrow™: Using Behavioral Economics to Increase Employee Saving." *Journal of Political Economy* 112 (S1): S164–S187.
- Thaler, Richard H., and Cass R. Sunstein.** 2008. *Nudge: Improving Decisions about Health, Wealth, and Happiness*. Yale University Press.
- Tversky, Amos, and Daniel Kahneman.** 1974. "Judgment under Uncertainty: Heuristics and Biases: Biases in Judgments Reveal Some Heuristics of Thinking under Uncertainty." *Science* 185 (4157): 1124–31.
- Vihriälä, Erkki.** 2025. "Intrahousehold Frictions, Anchoring, and the Credit Card Debt Puzzle." *Review of Economics and Statistics* 107 (2): 510–522.
- Wang, Jialan.** 2024. "To Pay or Autopay? Fintech Innovation and Credit Card Payments." NBER Working Paper 32332.
- Zinman, Jonathan.** 2015. "Household Debt: Facts, Puzzles, Theories, and Policies." *Annual Review of Economics* 7: 251–76.